

ETF5231 Group Assignment 2

Group7 GenieUS

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Question 1

What is your data about (no more than 50 words for each series)? Produce appropriate plots in order to become familiar with your data. Make sure you label your axes and plots appropriately. Comment on these. What do you see? (no more than 50 words per plot). (14 marks)

```
# Name and unit of data
sector_name <- meta %>%
  filter(`Student ID` == "Sector") %>%
  pull(`Group 7`)

data_unit <- meta %>%
  filter(`Student ID` == "Unit") %>%
  pull(`Group 7`)

cat("Sector: ", sector_name, "\n")
```

Sector: Turnover ; Australian Capital Territory ; Newspaper and book retailing ;

```
cat("Unit: ", data_unit, "\n")
```

Unit: \$ Millions

```
head(print_retail_act, 3)
```

```
# A tsibble: 3 x 2 [1M]
  Month      y
  <mtth> <dbl>
1 1982 Apr   2.3
2 1982 May   2.5
3 1982 Jun   2.3
```

```
tail(print_retail_act, 3)
```

```
# A tsibble: 3 x 2 [1M]
  Month      y
  <mtth> <dbl>
1 2023 Apr   1.8
2 2023 May   1.8
3 2023 Jun   1.6
```

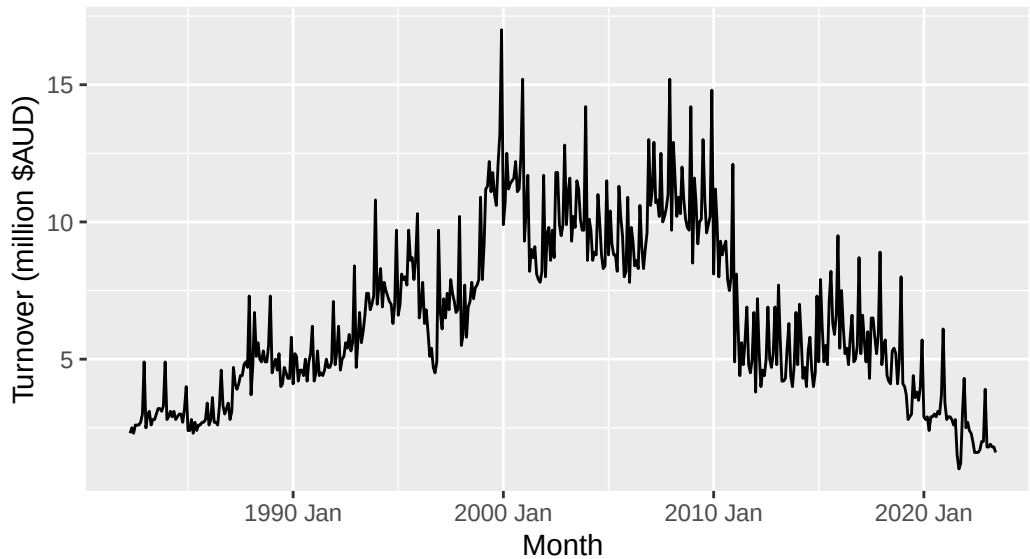
What is the data about

This series records monthly newspaper and book retail turnover in Australian Capital Territory, measured in millions of dollars, over the period April 1982 to June 2023.

Get to know the data

```
print_retail_act %>% autoplot(y) +
  labs(x = "Month",
       y = "Turnover (million $AUD)",
       title = "Newspaper and book retailing",
       subtitle = "Australian Capital Territory")
```

Newspaper and book retailing Australian Capital Territory

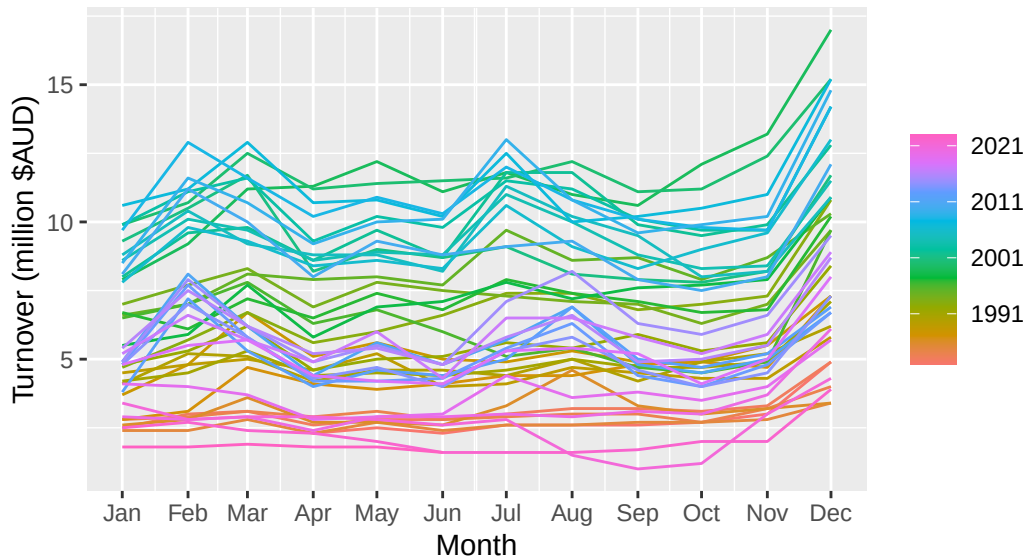


The plot shows clear monthly seasonality with regular peaks and drops. It also changes over time, rising in the earlier years and falling in the later years. The ups and downs look bigger when turnover is higher, so the variation does not seem constant.

```
print_retail_act %>% gg_season(y) +  
  labs(x = "Month",  
       y = "Turnover (million $AUD)",  
       title = "Newspaper and book retailing",  
       subtitle = "Australian Capital Territory")
```

Warning: `gg\_season()` was deprecated in feasts 0.4.2.
i Please use `ggtime::gg\_season()` instead.

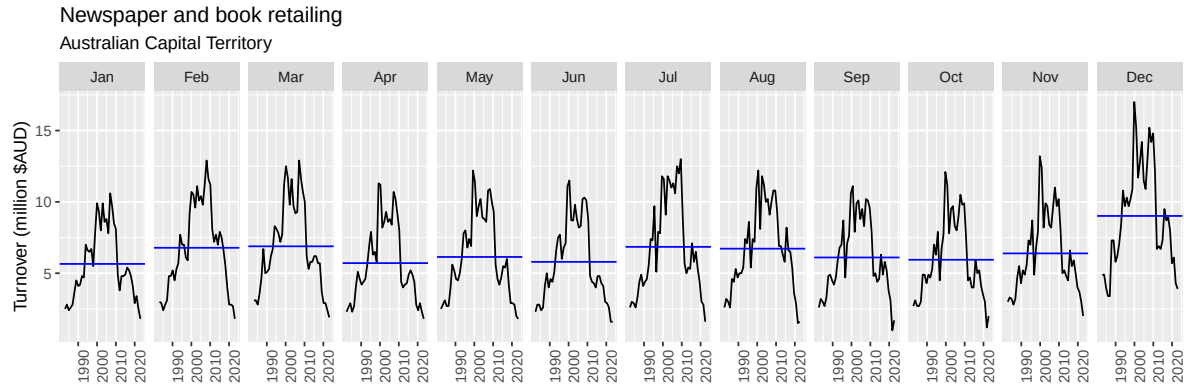
Newspaper and book retailing Australian Capital Territory



The seasonal plot shows turnover often rises in February and July and peaks in December. This may reflect back-to-school spending, mid-year sales or holidays, and Christmas shopping. Recent years are also lower than earlier years, suggesting a fall in overall turnover.

```
print_retail_act %>% gg_subseries(y) +  
  labs(x = "",  
        y = "Turnover (million $AUD)",  
        title = "Newspaper and book retailing",  
        subtitle = "Australian Capital Territory")
```

Warning: `gg\_subseries()` was deprecated in feasts 0.4.2.
i Please use `ggtime::gg\_subseries()` instead.



The subseries plot confirms clear seasonality. February–March and July–August generally have higher turnover than most months, while December has the highest average. It also shows that turnover in most months has fallen in recent years compared with earlier years.

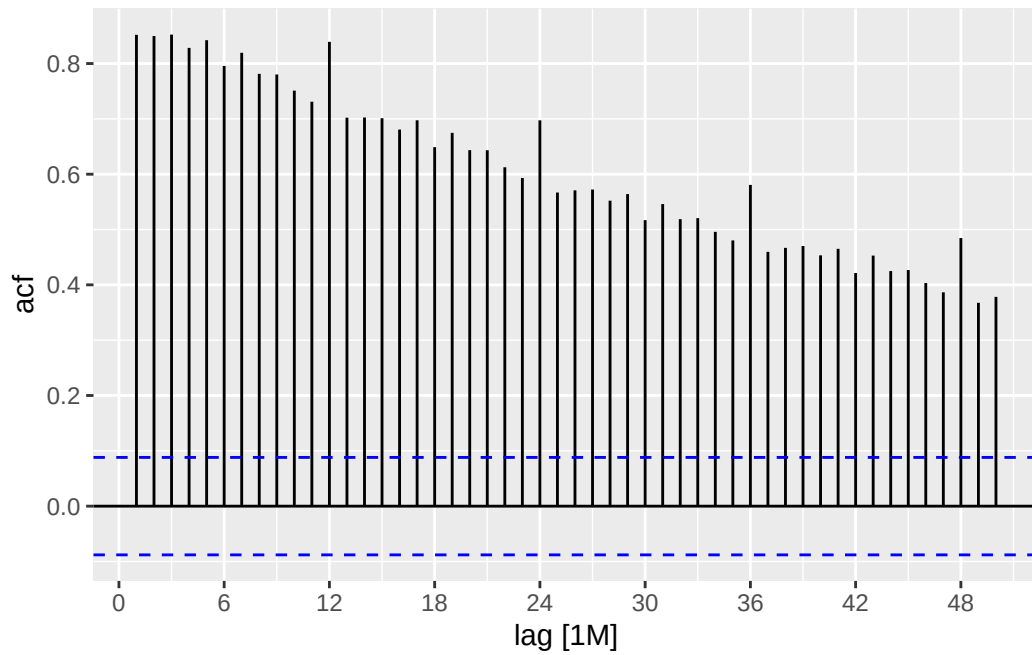
```
print_retail_act %>% gg_lag(y, lags=1:24, geom='point') +
  facet_wrap(~ .lag, ncol=6) +
  labs(y = "Turnover (million $AUD)",
       title = "Newspaper and book retailing",
       subtitle = "Australian Capital Territory")
```

Warning: `gg\_lag()` was deprecated in feasts 0.4.2.
i Please use `ggtime::gg\_lag()` instead.



The lag plots show an upward pattern rather than a random cloud. Points with high past turnover are generally associated with high current turnover, while low past values are associated with low current turnover. This suggests positive autocorrelation.

```
print_retail_act %>%
  ACF(y, lag_max = 50) %>%
  autoplot()
```



The ACF has large positive spikes across many lags and they remain above the significance limits. This shows the series is not white noise and that there is still strong dependence over time. There are also clear spikes at lags 12, 24, 36, and 48, which suggests a strong seasonal pattern with a period of 12. This indicates that observations are correlated with values from the same season in previous years.

Question 2

Would transforming your data be useful (no more than 50 words)? Choose a transformation and justify your choice (no more than 50 words). (5 marks)

```
lambda_tbl <- print_retail_act %>%
  features(y, guerrero)

lambda_tbl %>%
  mutate(lambda_guerrero = round(lambda_guerrero, 4)) %>%
  rename(`Guerrero lambda` = lambda_guerrero) %>%
  kable(caption = "Estimated transformation parameter from Guerrero's method")
```

Table 1: Estimated transformation parameter from Guerrero's method

Guerrero lambda
0.403

```
p1 <- print_retail_act %>%
  autoplot(y) +
  labs(
    title = "Original series",
    x = "Month",
    y = "Turnover ($ Millions)")

p2 <- print_retail_act %>%
  autoplot(box_cox(y, 0.403)) +
  labs(
    title = expression("Box-Cox transformed series (" * lambda == 0.403 * ")"),
    x = "Month",
    y = "Box-Cox(Turnover)")

p3 <- print_retail_act %>%
  autoplot(box_cox(y, 0.5)) +
  labs(
    title = "Square root transformed series",
    x = "Month",
    y = "Box-Cox(Turnover)")

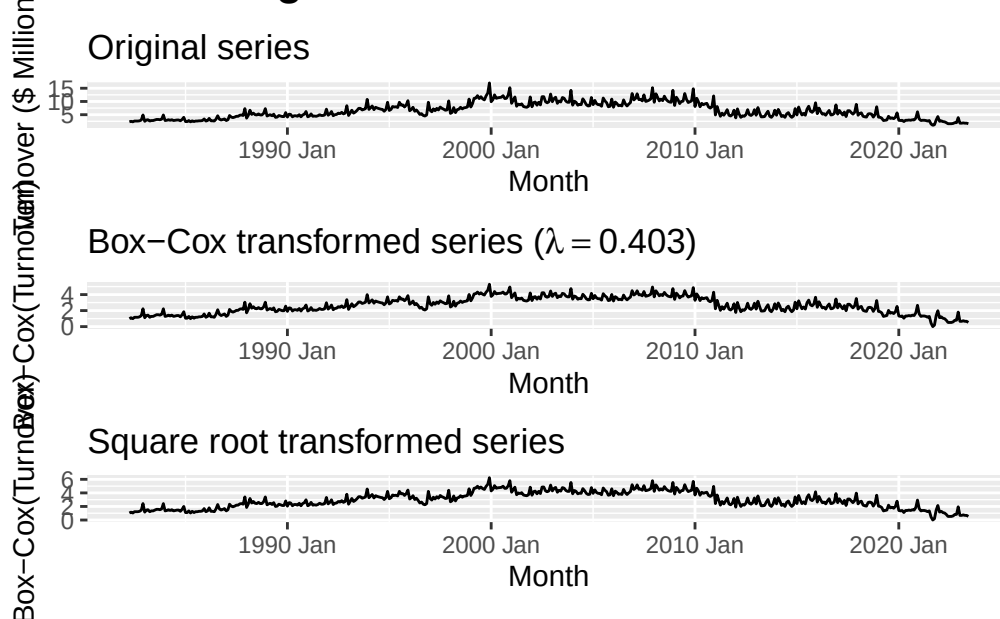
(p1 / p2 / p3) +
  plot_annotation(
```

```

title = "Comparison of the original series and Box-Cox transformed series",
theme = theme(
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16)))

```

Comparison of the original series and Box-Cox transformed series



Guerrero's method estimated $\lambda = 0.403$, so I compared the original series with Box-Cox transformations using $\lambda = 0.403$ and $\lambda = 0.5$; since the two transformed series looked very similar and $\lambda = 0.5$ is equivalent to a scaled and shifted square-root transformation, I use $\lambda = 0.5$ for easier interpretation.

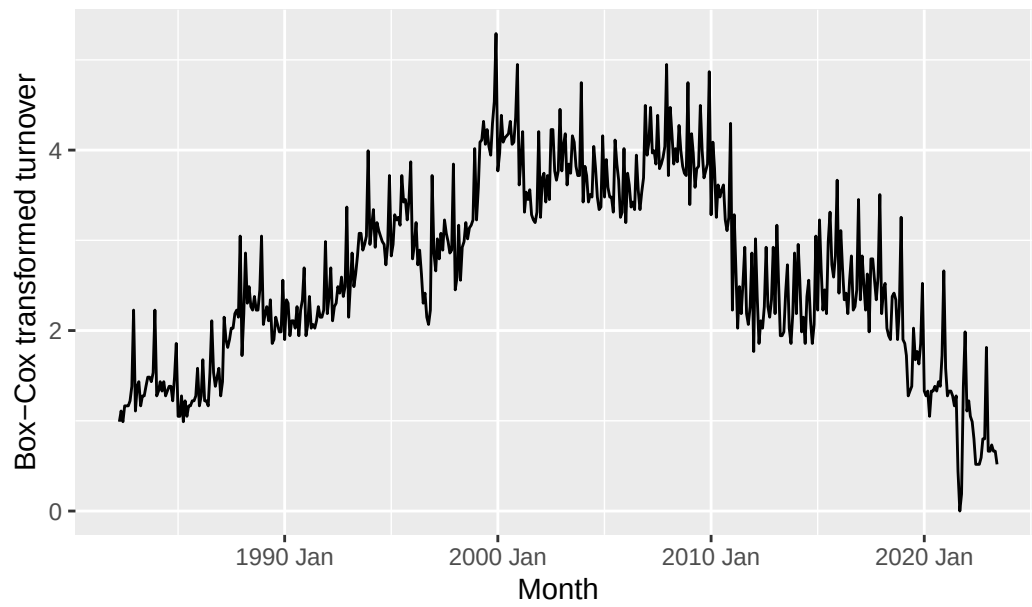
```

# Guerrero's method to estimate an appropriate lambda for the Box-Cox transformation.
lambda <- print_retail_act %>%
  features(y, features = guerrero) %>%
  pull(lambda_guerrero)

# Box-Cox plot
print_retail_act %>% autoplot(box_cox(y, lambda)) +
  labs(x = "Month",
       y = "Box-Cox transformed turnover",
       title = paste("Box-Cox transformed series (lambda =", round(lambda, 3), ")")
  )

```

Box-Cox transformed series (lambda = 0.403)



Question 3

Split your data into training and test sets. Leave the last two years' worth of observations as the test set. Apply all four benchmark methods on the training set. Generate forecasts for the test set and plot them on the same graph. Compare their forecasting performance on the test set. Which method would you choose as an appropriate benchmark? Justify your answer (no more than 100 words). (Hint: it will be useful to tabulate your results.) (6 marks)

Question 4

Can you construct better forecasting method(s) for your data? Give a brief description of these. Evaluate the method(s) over the test set and compare them to the benchmarks (no more than 50 words each). This question requires you to think about and only use tools you have acquired so far in this unit. Only materials from up to and including Chapter 5 (not Chapter 8) can be used. (Notice that more weight is now placed on this part of the assignment) (14 marks)

Question 5

Use time series cross-validation to select the best of the methods you have so far considered (you may only consider the most appropriate of the four benchmarks). Tabulate your results. How do these compare with the test set? Comment on the advantages and/or disadvantages of using this method (no more than 100 words). (10 marks)

Question 6

For the best method, do a residual analysis. Comment on these (no more than 150 words). (4 marks)

Question 7

Generate point forecasts for the next 2 years (future) from the method you have classified as best and plot them. Compare these visually and comment on the point and prediction intervals (no more than 100 words). (6 marks)

Question 8

Suppose that your group was a forecasting consulting group and you were working for a client. Give your group a name (add that to the subtitle above). Report your work here summarising and critically (think about assumption, limitations, extensions) evaluating the forecasts you are generating (no more than 200 words). (5 marks)